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6 October 1952

MILITARY SPECIFICATION

TEST STAND, HYDRAULIC SYSTEM, ELECTRIC MOTOR DRIVEN, TYPE D-6A

1. SCOPE

1.1 This specification covers one type of electric-motor-driven, trailer-mounted, aircraft hydraulic system test stand designated Type D-6A, and rated at 0 to 10 gpm, 0 to 3,000 psi or 0 to 5 gpm at 3,000 to 5,000 psi.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Federal

TT-C-595	Colors, (For) Ready-Mixed Paints
CC-M-641	Motors; Alternating-Current, Integral-Horsepower
GG-G-76	Gages, Pressure and Vacuum (For Air Ammonia, Steam, Oil, and Water)

Military

JAN-P-100	Packaging and Packing For Overseas Shipment, General Specification
JAN-P-104	Packaging and Packing For Overseas Shipment Crates, Sheathed, Wood, Nailed
MIL-P-116	Preservation, Methods Of
JAN-P-127	Packaging and Packing for Overseas Shipment-Tape, Adhesive, Pressure-Sensitive, Water-Resistant
MIL-T-712	Trucks, 2-1/2-Ton, 6 x 6, Series M44 and M133
MIL-T-1374	Truck, Utility, 1/4-Ton, 4 x 4, M38A1; Truck, Ambulance, Front Line, 1/4-Ton, 4 x 4, M-170
MIL-C-3769	Crates, Intermediate, Sheathed, Wood, Nailed (For Maximum Net Loads of 3000 Pounds)
MIL-E-4970	Environmental Testing, Ground Support Equipment, General Specification For

FSC 4920

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MIL-D-5028	Drawings and Data Lists; Preparation Of Manufacturers' (For Production Aircraft, Guided Missiles, Engines, Accessories, and Other Auxiliary Equipment)
MIL-H-5440	Hydraulic Systems; Design, Installation and Tests of Aircraft (General Specification For)
MIL-R-5520	Reservoirs; Hydraulic
MIL-O-5606	Oil; Hydraulic, Aircraft, Petroleum Base
MIL-C-5756	Cable and Wire, Power, Electric, Portable
MIL-I-6181	Interference Limits, Tests And Design Requirements, Aircraft Electrical And Electronic Equipment
MIL-W-8005	Wheels And Hubs, For Industrial Pneumatic Tires
MIL-A-8421	Air Transportability Requirements, Ground Support Equipment, General Specification For
MIL-T-9107	Test Reports; Preparation Of
MIL-L-10547	Liners, Case, Waterproof

STANDARDS

Military

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MS28741	Hose Assembly - Detachable End Fitting, Medium Pressure
MS33514	Fitting End, Standard Dimensions For Flareless Tube Connection and Gasket Seal
MS33515	Fitting End, Standard Dimensions For Bulkhead Flareless Tube Connections

Air Force-Navy Aeronautical

AN832	Union-Flared Tube 3/8 Bulkhead and Universal
AN924	Nut-Flared Tube, Bulkhead And Universal Fitting
AN929	Cap Assembly, Pressure Seal Flared Tube Fitting
AN6235	Filter Element - Hydraulic Replaceable Micronic Line Type
AN6236	Filter Element - Hydraulic Replaceable Micronic Reservoir Type
AN6249	Vavle - 3000 PSI Hydraulic Check
AN6292	Hose Assembly - Hydraulic and Pneumatic High Pressure (3000 PSI) Swivel Connections

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AND10050	Bosses, Standard Dimensions For Gasket Seal Straight Thread
AND10056	Fitting End, Standard Dimensions For Flared Tube Connection and Gasket Seal

DRAWINGS

Ordnance Corps

B7041347	Lunette Eye

Air Force

48B7385	Ring - Cargo Tie Down
48B7796	Ring Assembly - Tie Down, 10000 Lb
4936564	Ring - Eyebolt, 25000 Lb Cargo Tiedown
55C6182	Connectors - 440V/416V Power Cable

PUBLICATIONS

Air Force-Navy Aeronautical Bulletin

143	Specifications and Standards; Use of

Air Force Bulletin

F49	Codes for Ground Equipment Finishing Schemes

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 OTHER PUBLICATIONS.- The following documents form a part of this specification. Unless otherwise specified, the issue in effect on date of invitation for bids shall apply.

AMERICAN STANDARDS ASSOCIATION

Standard C-50	Rotating Electrical Machinery

(Copies of the ASA standard may be obtained from the American Standards Association, 70 East 45th Street, New York, New York.)

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION

Standard MG-1	Motors and Generators

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(Copies of the NEMA standard may be obtained from the National Electrical Manufacturers' Association, Engineering Department, 155 East 44th Street, New York, New York.)

NATIONAL BOARD OF FIRE UNDERWRITERS'

Pamphlet 70 National Electrical Code (Electric Wiring and Apparatus)

(Copies of the NBFU pamphlet may be obtained from the National Fire Protection Association, 60 Batterymarch Street, Boston 10, Massachusetts.)

3. REQUIREMENTS

3.1 PREPRODUCTION.- This specification makes provisions for preproduction testing.

3.2 COMPONENTS.- The stand shall consist of the following major components:

Description	See Requirement
a. Hydraulic system	3.9
b. Electrical system	3.10
c. Enclosure and trailer	3.11

3.3 GENERAL.- The requirements for materials, fluids, fittings, reservoir venting, tubing, and cleaning of parts and systems specified in MIL-H-5440 apply as requirements of this specification.

3.3.1 REQUIREMENTS CONFLICT.- Where the requirements of MIL-H-5440 conflict with the requirements of this specification, the requirements of this specification shall govern.

3.4 SELECTION OF SPECIFICATIONS AND STANDARDS.- Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with Bulletin 143, except as specified in 3.4.1 and 3.4.2.

3.4.1 COMMERCIAL PARTS.- Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, such as screws, bolts, nuts, and cotter pins, having suitable properties may be used provided:

a. They can be replaced by the standard parts (MS or AN) without alteration.

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b. The corresponding standard part numbers are referenced in the parts list and, if practical, on the contractor's drawings.

3.4.2 STANDARD PARTS.- With the exception specified in 3.4.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.4.3 All fittings and threaded bosses in all components shall be as specified in AND10050, AND10056, MS33514, and MS33515. Taper pipe threads shall not be used. All 5,000-psi pressure fittings shall be of the steel and bulkhead type. High-pressure fittings shall use a locknut with a backup gasket.

3.5 MATERIALS

3.5.1 PROTECTIVE TREATMENT.- When materials are used in the construction of the stand that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.6 DESIGN.- The stand shall be designed to be a self-contained, mobile testing unit providing not less than 10 gpm flow at 3,000 psi and 5 gpm flow at 5,000 psi with components arranged as shown on figure 1. Components shall be arranged for maximum ease of operation and maintenance consistent with the versatility and efficiency required to conform to the requirements of section 4. The stand shall be suitable for continuous full capacity operation from sea level up to and including 6,000 feet pressure altitude.

3.7 CONSTRUCTION.- The stand shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.7.1 ADJUSTMENTS AND REPAIRS.- The stand shall be constructed such that adjustments, repairs, and overhaul are readily accessible to and easily accomplished by the personnel of operating units and overhaul bases. The stand shall be constructed such that simple handtools may be used for maintenance and servicing and no special tools will be required. The stand shall be constructed to provide the maximum safety to personnel operating or maintaining the unit.

3.7.2 FORCES AND LOADINGS.- The stand structure, components, and mounting shall be constructed to withstand the gravitational forces and acceleration loadings specified in MIL-A-8421.

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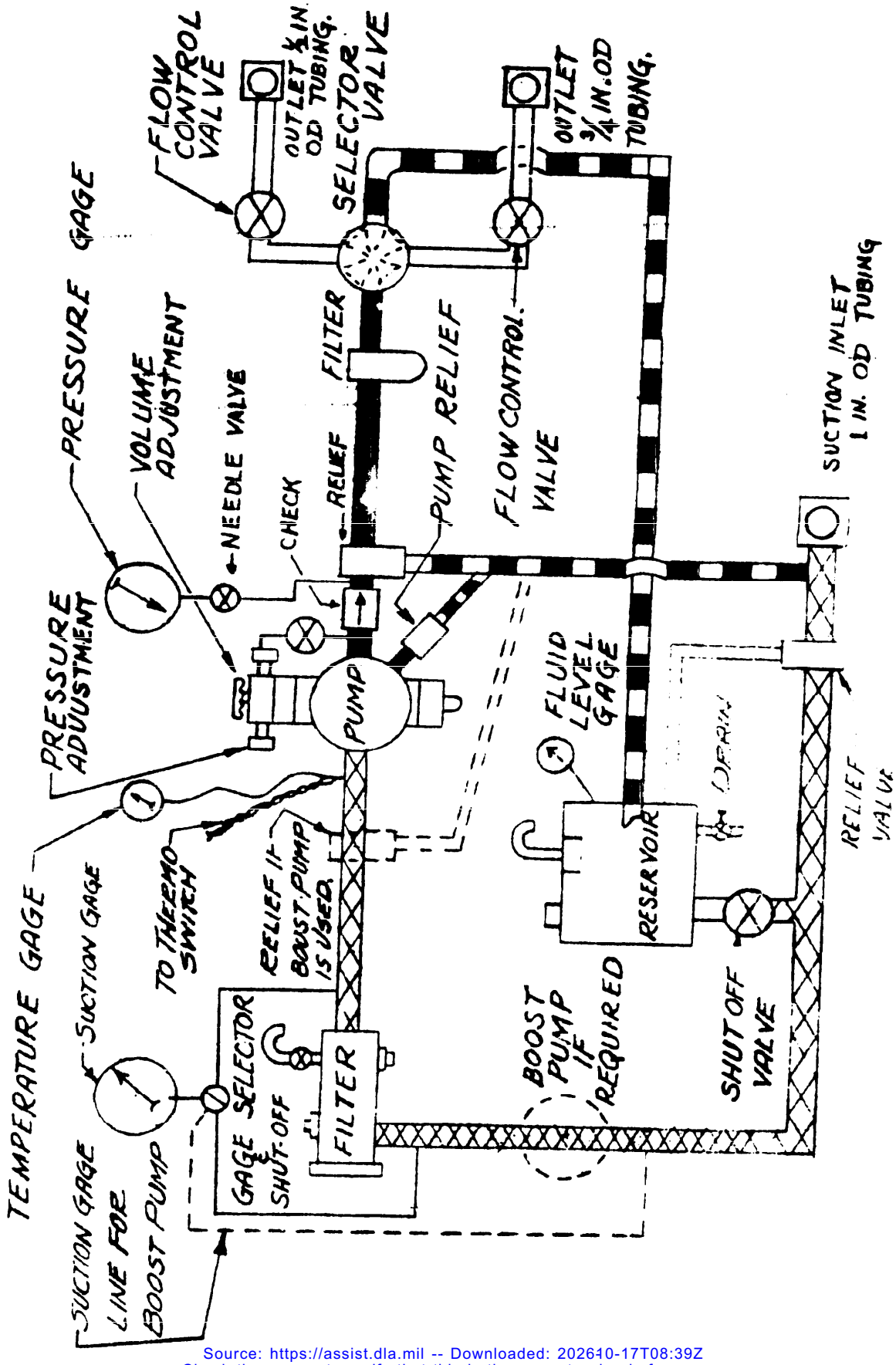


FIGURE 1 SCHEMATIC HYDRAULIC DIAGRAM

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3.8 PERFORMANCE.- The stand shall be capable of operating under and shall show satisfactory resistance to the following:

a. Operating temperatures ranging from -20° to +125°F and storage at temperatures ranging from -65° to +160°F

b. A relative humidity of 95 to 100 percent at an operating temperature of +125°F and a storage temperature of +160°F

c. Rainfall as encountered in any locale

d. Exposure to salt sea atmosphere

e. Self-induced vibration for all output conditions and vibration encountered in shipment

3.8.1 OPERATING POSITIONS.- The stand shall be capable of functioning satisfactorily when its normal operating plane is at an angle of 8-1/2° in any direction from horizontal.

3.8.2 VOLTAGE.- The stand shall be designed for continuous operation on 220V ± 10 percent and for conversion to operation on 440V ± 10 percent.

3.8.3 FREQUENCY.- The stand shall be designed for continuous operation on 60 cycles ±5 percent.

3.8.4 NOISE LEVEL.- The noise level of the stand shall not exceed the values in the frequency ranges specified in 4.4.9.

3.8.5 RADIO INTERFERENCE.- The stand shall meet the requirements of MIL-I-6181, 0.15 through 150 mc (radiated).

3.9 HYDRAULIC SYSTEM

3.9.1 HIGH-PRESSURE PUMP.- The high-pressure pump shall be a variable-stroke, variable-pressure, axial, piston-type pump with a pressure compensator and handwheel stroke control. The high-. pressure pump shall be of adequate rating to meet the requirements specified herein.

3.9.1.1 The pump shall not require more than 20 bhp from the drive motor when pumping hydraulic fluid conforming to MIL-O-5606 continuously at the rate of 10 gpm and raising the fluid pressure from 0 psi gage at 68°F to 3,000 psi gage.

3.9.1.2 The pressure and volume adjustments shall be capable of being adjusted to and fixed at any point within the range of from 800 to 5,000 psi pressure and 0 to 10 gpm flow.

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3.9.1.3 PRESSURE COMPENSATOR CONTROL.- The pump shall have an adjustable pressure compensator control capable of regulating the pressure at a constant value with the flow varying from 0 to 10 gpm. The adjustment range of the pressure compensator shall be from 800 to 5,000 psi. A pressure compensator shutoff valve shall be provided to permit turning off the pressure compensator to facilitate adjustment of the relief valve to any valve within the pressure range from 400 to 5,000 psi.

3.9.1.4 A relief valve shall be provided on the inlet side of the pump to return hydraulic fluid to the reservoir. The setting of the relief valve shall be that recommended by the pump manufacturer and shall be capable of protecting the pump case in the event the return system from the aircraft is pressurized.

3.9.1.5 The high-pressure pump shall be capable of maintaining rated output of 10 gpm at 3,000 psi or 5 gpm at 5,000 psi at the same time that the inlet fluid pressure to the stand is varied throughout the range of 16 inches Hg absolute to 110 psig. If necessary, a boost pump may be used. If a boost pump is used, the stand shall be capable of full rated output while the inlet pressure to the boost pump is 18 inches Hg vacuum. If a boost pump is provided, a relief valve shall be provided to limit the pressure between the boost pump and high-pressure pump to a value not less than 10 psi nor more than 110 psig.

3.9.2 HIGH-PRESSURE FILTER.- A high-pressure filter employing a relief bypass valve and one or more AN6235-4A micronic elements shall be provided in the pressure line.

3.9.3 LOW-PRESSURE FILTER.- A suction line filter shall be provided and located as shown on figure 1. The filter shall utilize a minimum of one replaceable AN6236-3 or two replaceable AN6236-2 filter elements. Filling and draining plugs and air bleed valves shall be furnished and sufficient clearance space allowed around each item for draining, bleeding, filling, and replacing the filter elements.

3.9.4 RELIEF VALVES.- A high-pressure relief valve shall be adjustable by hand from 400 to 5,000 psig. A low-pressure relief valve, if provided, shall be adjusted to a fixed pressure of not less than 20 nor more than 110 psi as recommended by the high-pressure pump manufacturer. Relief valves shall be located in the system as shown on figure 1. The adjustable high-pressure relief valve shall be mounted on the control panel and shall incorporate a locking device so that vibrations cannot change the setting.

3.9.4.1 BYPASS VALVE.- A valve suitable for 5,000 psi shall be provided and installed to bypass the high-pressure pump outlet to the return line to facilitate starting the stand at zero pressure.

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3.9.5 RESERVOIR.- A hydraulic-fluid reservoir shall be provided and mounted within the stand enclosure. The reservoir shall have adequate capacity to supply at least 20 gallons of fluid to the system when inclined at 8-1/2° from vertical, in any direction, plus additional space for a 3-gallon expansion above the normal full level, and at least a 2-gallon space below the outlet to the system for the collection of moisture and impurities.

3.9.5.1 The reservoir shall conform to the requirements of MIL-R-5520 for obstruction to flow, structural design, shell, seams, structural strength, baffles, fluid connections, sump, drain, vent, vent filter, vent line, return line inlet, filler unit, finish, and filling time. Adequate clearance shall be provided around the filler opening to allow rapid filling from 1-gallon cans while the stand is in operation.

3.9.5.2 ACCESS PLATES.- One or more access plates, at least 8 by 8 inches, shall be provided to allow visual inspection and manual cleaning of the interior of the reservoir.

3.9.5.3 DRAIN VALVE.- A drain valve shall be provided for completely draining the reservoir. Necessary clearance shall be provided around the drain outlet to permit rapid draining into 1-gallon containers. A 1-inch, or larger, reservoir shutoff valve shall be provided and installed on the control panel and as shown on figure 1.

3.9.6 PRESSURE OUTLETS.- Two pressure outlets shall be provided as shown on figure 1. One outlet shall be 1/2-inch tube size and the other outlet shall be 3/4-inch tube size. Each outlet shall be equipped with the corresponding size Aeroquip Part No. 155-S4, or equal, and interchangeable coupling half for connection to its respective high-pressure hose. Each outlet shall be provided with an Aeroquip Part No. 155-S7, or equal, and interchangeable dust cap. The dust caps shall be permanently attached to the stand by means of a metal chain to prevent loss. The location shall not allow leakage from the connection to fall on the trailer tires nor within the stand enclosure.

3.9.6.1 PRESSURE CONNECTIONS.- All valves, connections, and tubing used from the high-pressure pump discharge to the end of the 20-foot pressure hose shall be of a size that will not allow pressure drop to exceed 20 psi. All connections and fittings on the high-pressure side shall be not less than 3/4-inch tubing size. Tubing shall be used on all high-pressure lines within the stand.

3.9.7 SELECTOR VALVE.- A four-way closed-center, two-position, rotary-type selector valve, Barksdale Valve No. 20HP6HC3, or equal, designed for 5,000 psi working pressure shall be provided and

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installed as shown on figure 1. The selector valve shall utilize O-rings to prevent external leakage. The selector valve shall be installed in the stand system to direct pressure to the 1/2-inch outlet and allow fluid to return through the 3/4-inch outlet to the reservoir while in one position, and to direct pressure to the 3/4-inch outlet and allow return through the 1/2-inch outlet to the reservoir while in the second position.

3.9.8 SUCTION INLET.- The suction inlet shall be provided with one Aeroquip Part No. 155-S4-16 coupling half with Aeroquip Part No. 155-S7-16 dust cap, or equal, and interchangeable part. The dust cap shall be permanently attached to the stand by means of a metal chain to prevent loss. The location shall not allow leakage from the connector to fall on the trailer tires nor within the stand enclosure.

3.9.8.1 LOW-PRESSURE COMPONENTS AND CONNECTIONS.- The size of all connections used in the suction side of the stand hydraulic system shall be such as to allow not more than a 4 psi pressure drop at 10 gpm with a fluid temperature of 68°F from the aircraft outlet through 10 feet of hose to the suction side of the high-pressure pump, or boost pump if used. Suction lines and components shall be reduced to not less than 1-inch tubing size.

3.9.9 FLOW CONTROL VALVES.- Two valves of 1/2 inch and 3/4 inch or larger shall control or stop the flow to each pressure outlet. The valves shall be suitable for 10 gpm flow and 5,000 psi pressure. The valves shall accurately meter the flow throughout the volume range of 0 to 10 gpm and shall be pressure balanced so that a force of 40 in. lb is the maximum required turning force at 3,000 psi. The valves shall be capable of going from the closed position to the open position in two full turns of the handle. The valves shall be capable of controlling the flow within ±1/4 gpm when flow rates are below 5 gpm, and ±1 gpm with flow rates of from 5 to 10 gpm.

3.9.9.1 BYPASS VALVE.- A bypass valve of 1/2 inch tube size or larger which meets the requirements of 3.9.9 or a pilot valve to dump the relief valve shall be provided and mounted on the control panel.

3.9.10 PRESSURE CHECK VALVE.- A pressure check valve which is dimensionally interchangeable with an AN6249-12 pressure check valve but suitable for operation at 5,000 psi shall be located as shown on figure 1 to prevent reverse flow to the high-pressure pump.

3.9.11 FLOW INDICATOR.- A direct reading means shall be provided to determine actual flow through the high-pressure pump within ±1/2 gpm

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at all conditions of flow and pressure. This shall be accomplished by means of a calibrated indicator connected to the pump assembly. The flow measuring device shall indicate actual flow during reduced flow condition due to compensator action. If a flowmeter independent of the high-pressure pump is used, the flow adjustment on the pump shall be calibrated in 1-gpm increments to allow approximately maximum flow adjustment before starting test run.

3.9.12 HIGH-PRESSURE GAGE.- A flush-mounted, high-pressure gage shall be provided and shall have a 4-1/2-inch diameter dial and a steel-alloy bourdon tube. The gage shall be accurate to within 1 percent, or less, of full scale and shall conform to GG-G-76, except, where conflicts exist, the provisions of this specification shall govern. The gage shall provide a range of from 0 to 6,000 psi and shall incorporate a means to restore the accuracy of the gage under all service conditions. The gage shall be of the recess type provided with a mounting flange at the front of the gage. The mounting flange and dial shall not extend more than 5/8 inch beyond the external surface of the panel. The gage shall incorporate a safety blowout device. The safety blowout device shall automatically relieve the gage case pressure should it reach 1/2 psi due to gage failure. The pointer shall be adjustable without removing the gage from the panel. Good quality stainless steel shall be required on the gage.

3.9.12.1 MOUNTING.- The gage shall be mounted by screws and tapped panel holes to permit removal by one man from in front of the panel.

3.9.12.2 CONNECTION.- The gages shall be connected to the panel with flexible AN- or MS-type hose, or commercial equivalent, of sufficient length to permit removal and installation from the front of the panel.

3.9.12.3 GAGE LINE.- The high-pressure gage line shall be provided with a needle shutoff valve, an accumulator-type pulsation dampener, and gage test fittings. The pulsation dampener shall prevent gage damage from pressure pulsations and pressure shock.

3.9.12.4 GAGE TEST FITTINGS.- The following fittings shall be mounted on the control panel as shown on figure 1, adjacent to the appropriate gage. The panel shall be marked to indicate that the fittings are gage test connections.

a. An AN832-4D nipple

b. An AN924-4D nut

c. An AN929-4 cap

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3.9.13 LOW-PRESSURE GAGE.- A flush-mounted compound gage shall be provided and shall have corrosion-resistant, steel bourdon tubes and a dial of 4 inches diameter or larger. The gage shall be accurate within 1 percent of full scale range and shall conform to GG-G-76, except that, where conflicts exist, the provisions of this specification shall govern. The gage shall provide a range of 30 inches Hg vacuum to 150 psi. The gage shall be red sectored below 20 inches Hg vacuum and above 130 psi. The gage movements shall be constructed of stainless steel and shall be suitable for operation from -40° to +125°F.

3.9.13.1 A selector valve shall be provided to select the gage pressures before and after the low-pressure filter and to shut off the gage entirely as shown on figure 1.

3.9.13.2 In the event a boost pump is used to charge the high-pressure pump, the gage specified in 3.9.13 shall be a combination gage with range from 30 inches Hg vacuum to a pressure twice the boost pump relief valve setting. The selector valve specified in 3.9.13.1 shall provide an additional position to read, on the gage, the suction pressure to the boost pump.

3.9.14 TEMPERATURE GAGE.- A remote reading temperature gage shall be provided and shall indicate the temperature of the fluid entering the high-pressure pump. The range of the temperature gage shall be -20° to approximately 250°F and shall provide an accuracy of ±5°F, or better. The gage shall have a 2-1/2-inch, or larger, diameter dial and shall be flush mounted. The dial face of the temperature gage shall be marked with a red sector at 160°F and beyond as a warning.

3.9.15 FLUID LEVEL INDICATOR.- An electric-type fluid level indicator shall be provided to indicate the quantity of available fluid in the reservoir, accurate within ±2 gallons of actual quantity, and shall be clearly visible to the operator standing at the control panel.

3.9.16 HOSE ASSEMBLIES.- The following commercial hydraulic hose assemblies shall be provided:

a. Two each 1/2-inch hose assemblies 10 feet long, similar to AN6292-8-120 hose assemblies, except impulse requirements are waived and it need only be suitable for -20°F operation. The hose working pressure shall be 5,000 psi.

b. One each 3/4-inch hose assembly 10 feet long, similar to the AN6292-12-120 hose assembly, except impulse requirements are waived and it need only be suitable for -20°F operation. The hose working pressure shall be 5,000 psi.

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c. Two each 1-inch hose assemblies 10 feet long, similar to MS28741-16-120 hose assemblies, except impulse requirements are waived and it need only be suitable for -20°F operation. The hose working pressure shall be from 25 inches Hg vacuum to 800 psi.

3.9.17 HOSE ACCESSORIES.- Two Aeroquip Part No. 150-S1-8D coupling halves with 155-25-8 nuts and 155-S9-8 dust plugs, or equal, and interchangeable parts, two Aeroquip Part No. 150-S1-12D coupling halves with 155-25-12 nuts and 155-S9-12 dust plugs, or equal, and interchangeable parts, and two Aeroquip Part No. 140-S1-16D coupling halves with 145-25-16 nuts and 145-S9-16 dust plugs, or equal, and interchangeable parts shall be furnished and installed in each end of its respective hose specified in 3.9.16. Dust plugs shall be securely attached to the hose by means of Aeroquip Part No. 5100-S1, or equal, chain assemblies.

3.10 ELECTRICAL SYSTEM.- All electrical equipment and wiring shall conform to Pamphlet 70. All electrical equipment less than 18 inches from the ground shall be explosionproof for class I, division 2 location in compliance with section 5115 of Pamphlet 70. The electrical system shall be designed for operation on 220V, 3-phase, 60-cycle power supply and shall be designed for conversion to 440V, 3-phase, 50-cycle, without replacing any major component. The voltage tolerance shall be ±10 percent and the frequency ±5 percent.

3.10.1 PUMP MOTOR.- A design B or C, code E or F, ball bearing, squirrel cage induction motor conforming to Standard MG-1 shall be provided for driving the high- and low-pressure pumps. The pump motor shall conform to CC-M-641 and shall be of adequate rating to operate the stand continuously under any conditions specified in 3.8, at maximum capacity, without exceeding the NEMA recommended temperature limits for the class of insulating materials used. The manufacturer's recommended full load current shall not be exceeded during operation. The motor shall be suitable for operation at a pressure altitude of 6,000 feet.

3.10.2 PUMP MOTOR CONTROL.- The stand shall be provided with a remote-operated, magnetic motor control system in accordance with figure 2, providing motor overload protection and circuit protection. The remote-operated control shall be powered by 110V, obtained by reducing the input voltage with a transformer. Voltages greater than 120V shall not be used on the control panel.

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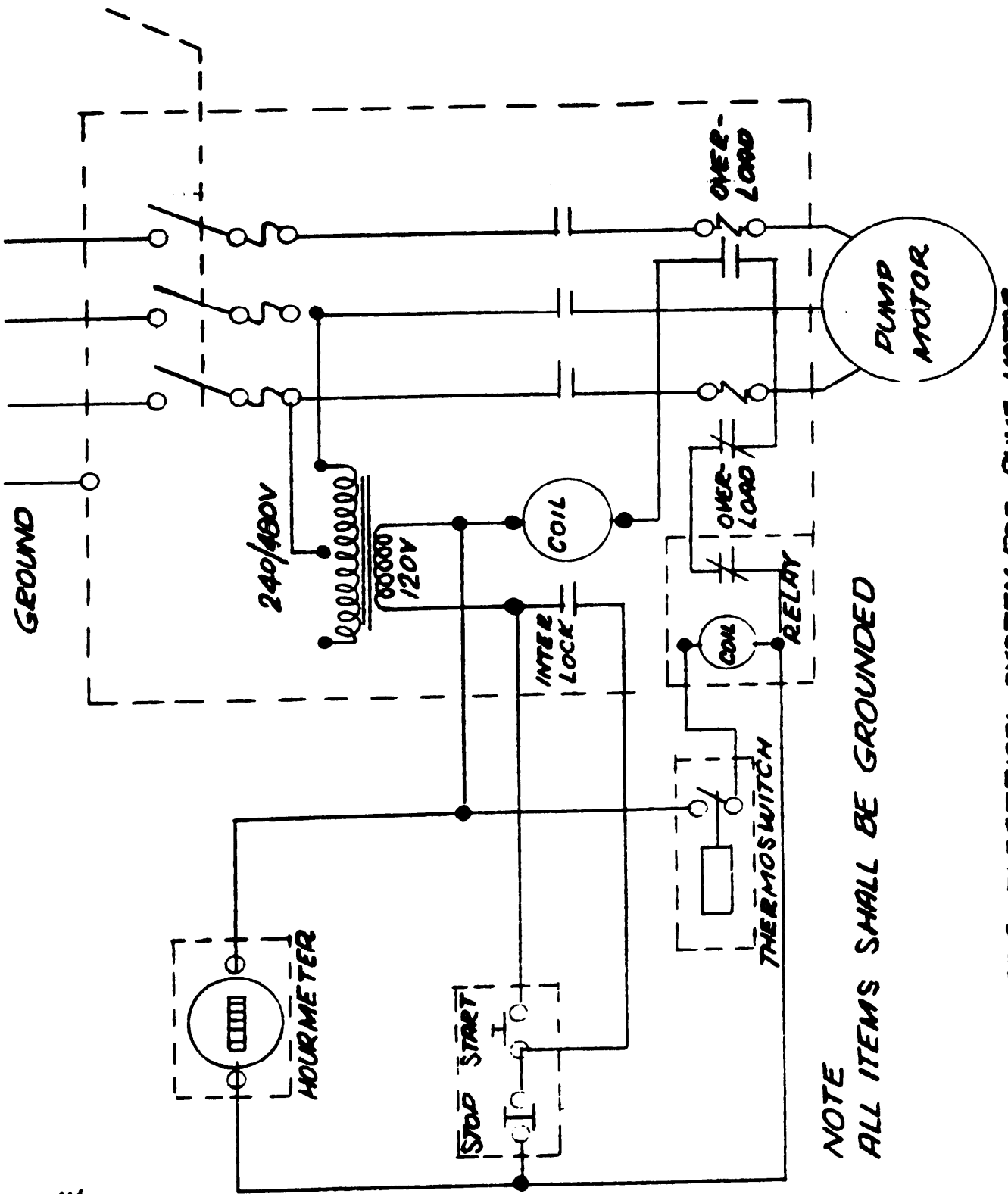


FIGURE 2. ELECTRICAL SYSTEM FOR PUMP MOTOR

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3.10.2.1 STARTER AND STARTER CASE.- A starter of adequate capacity to withstand openings under starting surges shall be provided. A starter case of sufficient size to enclose all remote controls (contactors, overload devices, transformers, starters) for all electric motors shall be furnished. The starter shall include two overload relays for motor protection which can be manually reset without opening the starter case. A 240/480 to 120V control transformer shall be furnished in the starter enclosure. Complete instructions shall be furnished on a permanent-type nameplate fastened inside the starting box giving all details and directions for converting the stand from operation on 240 to 480V or vice versa.

3.10.2.2 PUSHBUTTON.- One start-stop, momentary-contact, low-voltage-protection pushbutton shall be provided and mounted on the instrument panel.

3.10.2.3 CIRCUIT BREAKER.- An enclosed circuit breaker, rated in compliance with Pamphlet 70, shall be furnished. The circuit breaker may be combined in the same enclosure with the starter or provided as a separate unit.

3.10.2.4 WIRING.- All wiring between electrical components shall be in accordance with Pamphlet 70.

3.10.2.5 HOURMETER.- A counter-type hourmeter having a range of from 0 to 9,999 hours shall be provided to record the hours of operation of the pump motor.

3.10.2.6 ELECTRIC CABLE AND PLUG.- The alternating-current input cable shall be an extra-flexible, six-conductor cable in accordance with MIL-C-5756, 50 feet in length. The cable shall utilize three AWG No. 0 conductors, one conductor not smaller than No. 8 and two conductors not smaller than No. 12. The cable shall be provided with a molded-rubber, polarized, watertight plug acceptable to the procuring activity. The plug shall be in accordance with Drawing 55C6182. The cable shall be securely attached to the stand to take all strain from the terminals.

3.10.2.7 TEMPERATURE CONTROL DEVICE.- A thermoswitch and normally closed relay shall be provided as shown on figures 1 and 2 to shut down the stand when the hydraulic fluid temperature reaches 160°F.

3.10.2.8 REVERSE ROTATION PROTECTION.- A mechanical or electrical device shall be provided which will prevent operation of the pump motor in the opposite direction of rotation from that required for the pumps, in the event of incorrect phasing.

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3.11 ENCLOSURE AND TRAILER.- The components of the stand shall be mounted and assembled on a trailer and covered by a weather-resistant housing.

3.11.1 TRAILER.- The trailer shall be of the 4-wheel, 2-wheel-steerable type.

3.11.1.1 WHEELS, TIRES, AND HUBS.- The trailer shall be mounted on wheels, hubs, and tires, size 6:00 by 9, in accordance with MIL-W-8005. The tires shall be of 6-ply rating.

3.11.1.2 TRAILER FRAME.- The trailer frame shall be of welded steel construction and shall be from 36 to 40 inches wide and from 48 to 60 inches long. The ratio of width divided by length shall not exceed three-fourths.

3.11.1.3 PARKING BRAKES.- The trailer parking brakes shall hold the stand in a fixed position during all operating conditions and positions specified herein. The brakes shall be of the internal expanding type capable of locking the rear wheels to cause the tires to skid on dry, rough concrete when sufficient force to move the trailer is applied. Brakes for both rear wheels shall be interconnected and shall operate from one handle.

3.11.1.3.1 The parking brake control device shall be appropriately marked with operating instructions for release and application of the brakes. The marking shall be 1/4-inch black letters and shall be visible to the operator.

3.11.1.4 SPRINGS.- Springs shall be furnished on the trailer to provide good riding qualities, thereby precluding damage to the mounted equipment. The springs shall be of such construction that not more than 2 inches is added to the frame height for spring deflection.

3.11.1.5 STEERING.- Knuckle-type steering, incorporating tie rods and kingpins, shall be provided on the trailer to provide a high degree of maneuverability. Adequate means for lubrication shall be provided. The cramping angle shall be not less than 40°.

3.11.1.6 The trailer shall be furnished with a hinged drawbar assembly suitable for towing the stand at 20 mph behind a standard military 2-1/2-ton, 6 by 6, Type M35 truck conforming to MIL-T-712. The drawbar shall have a hollow, lightweight lunette eye conforming to Drawing B7041347 with a 3-inch inside diameter and a 1-5/8-inch diameter cross section.

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3.11.1.7 The stand shall be capable of executing full turns to the right and to the left sides while being towed at a speed of 10 mph by a military 4 by 4, 1/4-ton, Type M38 truck conforming to MIL-T-1374 and shall not capsize or skid during these turns on dry concrete or macadam. The turning radius shall be the minimum obtainable from the type towing vehicle specified.

3.11.1.8 The trailer shall be capable of supporting 800 pounds in addition to the stand components, instrument panel, and housing.

3.11.1.9 Provision shall be made on the stand trailer for attaching four or more lifting and tiedown fittings in accordance with Drawing 4887385, 4887796, or 4936564, capable of withstanding the loadings specified in 3.7.2.

3.11.1.10 The fully loaded stand trailer shall provide at least 6 inches road clearance. The tread width shall be 49 inches.

3.11.2 CONTROL PANEL.- A control panel shall be provided for mounting all meters, valves, indicators, gages, switches, and controls furnished with the stand. The control panel shall be constructed of No. 12(0.1046 inch) U. S. gage, or thicker, sheet steel and supported by a welded-steel, self-supporting framework. The control panel frame shall be bolted or welded to the trailer bed and positioned so that wheels, tow bars, et cetera, will not interfere with the operator. The control panel shall be located at sufficient height above the ground and inclined at sufficient angle, 10° minimum, to facilitate observation and operation of all items by an operator 5 feet 10 inches tall. Instruments subject to errors from vibration shall be shock mounted. All items mounted on the control panel shall be identified as to use and cautions by etched or engraved metal or plastic name-plates permanently attached to the panel.

3.11.2.1 A metal or plastic, stamped or engraved, permanent-type caution sign displaying the following warning in 1/4-inch letters, or larger, shall be provided and securely attached in a conspicuous place on the instrument panel: CAPACITY 10 GPM AT 3,000 PSI, AT PRESSURE FROM 3,000 PSI TO 5,000 PSI, REDUCE FLOW TO 5 GPM, OR LESS.

3.11.2.2 Two caution signs of the same material, except the letters shall be 1 inch or larger, displaying the following warning shall be provided and securely attached in a conspicuous place on the outside surface of both sides of the enclosure:

WARNING - KEEP 10 FEET CLEAR OF AIRCRAFT
ENGINES AND FUEL TANK AREAS

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3.11.2.3 A 4- by 6-inch, engraved, permanent-type diagram plate arranged similar to figure 1 to show the hydraulic circuit of the stand shall be securely fastened to the center of the instrument panel.

3.11.3 HOUSING.- The stand components and control panel shall be completely enclosed and protected by a weather-resistant housing. The housing shall consist of a welded-steel framework covered with No. 16 U. S. gage (0.0598 inch) sheet steel, or heavier, with doors and adjustable louvers as required. The housing shall be capable of supporting a 200-pound weight distributed evenly over an area 4-1/2 by 5 inches and applied in any location on the top of the stand. The housing shall be adequately braced to prevent distortion or excessive vibration during handling and operation of the stand or handling of the housing while removed from the stand. The stand shall be designed so that water or hydraulic fluid spilled any place within the housing will drain to the exterior and shall not collect within the enclosure. Metal hooks shall be furnished on the exterior of the housing to permit temporary storage of the pressure hose. These hooks shall be designed to permit flat positioning against the housing when not in use.

3.11.3.1 DOORS.- The housing shall be provided with sufficient access and ventilating doors and adjustable louvers to permit access for operating and servicing and to ventilate the stand. The doors and louvers in the housing shall be free from binding and easy to operate with one hand. Provisions shall be incorporated to secure the doors rigidly in the full open or full closed position.

3.11.3.2 The housing shall be easily removable by actuating not more than 10 quick-acting disconnecting fasteners and lifting or sliding the entire housing off the remainder of the stand assembly. Handles shall be provided for this purpose.

3.11.4 COMPARTMENTS.- A handbook compartment, 2 by 9 by 12 inches, or larger, shall be provided and permanently and securely attached within the housing. Provisions shall be made for storage of the hose assemblies and cable specified in 3.9.16 and 3.10.2.6 within the outline of the housing. The compartments may be constructed on the trailer bed or may be fastened to the housing assembly.

3.12 INTERCHANGEABILITY.- All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-D-5028 shall govern changes in the manufacturer's part numbers.

3.13 DIMENSIONS.- The length and width of the stand and housing shall not extend beyond the outline of the wheels at any point. The overall height of the stand shall be held to a minimum consistent with good design practices, but shall not exceed 32 inches above the top of the tires of the trailer.

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3.14 WEIGHT.- The weight of the stand, including the trailer, shall not exceed 2,500 pounds.

3.15 FINISHES AND PROTECTIVE COATINGS

3.15.1 EXTERIOR SURFACES.- All exterior surfaces of the stand shall be finished in accordance with Bulletin F49, code F49-4. The color shall be yellow, color 1310.

3.15.2 INTERIOR SURFACES.- All interior surfaces of the stand shall be finished in accordance with Bulletin F49, code F49-4. The color shall be gray, color 1625, of TT-C-595.

3.15.3 All steel surfaces to be painted shall be thoroughly cleaned of grease, rust, loose mill scale, dirt, and other foreign materials that might cause inferior bonding of the primer. The cleaned metal surfaces shall be given a chemically treated, nonmetallic coating, intimately attached to the metal, which retards corrosion and provides greater adhesion of the applied primer. The treated surfaces shall be given one or more prime coats and two or more paint coats, as required, to withstand the environmental tests and operating tests specified herein without cracking, peeling, or permitting rust formation.

3.16 IDENTIFICATION OF PRODUCT.- Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130. The following special characteristic marking shall appear on the nameplate:

Rating 0 to 10 GPM, 0 to 3000 PSI or
0 to 5 GPM at 3000 to 5000 PSI
(220V, 3 Phase, 60 Cycle _____ AMPS, AC)
(Can be converted to 440V, 3 PH, 60 Cycle)
Towing Speed - 20 MPH MAX.

3.17 WORKMANSHIP

3.17.1 GENERAL.- The stand, including all parts and accessories shall be fabricated and finished in a workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, wiring, and riveting, alignment of parts and tightness of assembly screws, and bolts; et cetera.

3.17.2 RIVETING.- Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

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3.17.3 CLEANING.- The stand shall be thoroughly cleaned, and loose, spattered, or excess solder, metal chips, and other foreign material shall be removed during and after final assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 CLASSIFICATION OF TESTS.- The inspection and testing of the stand shall be classified as follows:

a. Acceptance tests . . . . . See 4.2

b. Preproduction testing. . . . . See 4.5

4.2 ACCEPTANCE TESTS.- The acceptance tests shall consist of the following:

a. Individual test

b. Sampling plan and tests

4.2.1 INDIVIDUAL TESTS.- Each stand shall be subjected to the following tests as described under 4.4:

a. Examination of product (See 4.4.1)

b. Operational check (See 4.4.2)

4.2.2 SAMPLING PLAN AND TESTS

4.2.2.1 LOT.- A lot shall consist of stands manufactured under essentially the same conditions and submitted for inspection at substantially the same time.

4.2.2.2 SAMPLING PLAN.- The following number of sample stands shall be selected at random from each lot and subjected to the test listed below and described under 4.4:

No. of stands in lot	No. of sample stands
1 - 50	1
51 - 150	2
Over - 150	3

a. Semi-endurance test (See 4.4.3)

4.2.2.2.1 REJECTION AND RETEST.- When one or more stands from a lot fail to meet the specification, acceptance of all stands in the lot will be withheld until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

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flow control handwheel settings shall be accurate within ±1/2 gallon of actual flow, and reservoir fluid level gage shall be checked for accuracy of ±2 gallons throughout its range.

4.4.2.1 The operation of the stand shall be checked at 1,500 and 3,000 psi at each rate of flow of 5 and 10 gpm and with compensation set at 1,500 and 3,000 psi respectively. The hydraulic pump shall put out full 5 and 10 gpm flow at all loading pressures below the 1,500 and 3,000 psi respectively, but flow shall fall to nearly zero as soon as the loading pressure reaches the compensation pressure setting.

4.4.2.2 The temperature of the fluid entering the high-pressure pump on the stand shall be raised above 157°F and the thermoswitch shall shut off the pump motor before the fluid temperature reaches 162°F, but not less than 158°F.

4.4.3 SEMI-ENDURANCE.- The stands shall be subjected to 20 percent of the endurance test time specified in 4.4.6. At the completion of this test, the stands shall satisfactorily pass the tests specified in 4.4.1 and 4.4.2.

4.4.4 INITIAL DISASSEMBLY INSPECTION.- Each stand submitted for preproduction tests shall be disassembled to the maximum practical extent and completely inspected, the breakdown weights determined, and measurements made which may be used to determine wear. All gages, meters, and instruments shall be laboratory checked against certified calibrating equipment, and failure to meet the accuracy requirements specified in section 3, for any item, shall be cause for rejection. All requirements of section 3 which may be determined by measurements or inspection shall be checked for compliance, and failure to meet the requirements of section 3 shall be cause for rejection. Following this inspection, the stand shall be reassembled and subjected to the remainder of the preproduction tests.

4.4.5 ENVIRONMENTAL TESTS.- The stand shall be subjected to the following environmental tests conducted in accordance with the specified procedures of MIL-E-4970. Unless otherwise specified, upon completion of each test, the stand shall be subjected to the tests specified in 4.4.2 and 4.4.2.1. Failure of the stand to operate as required during or after each exposure, failure of any item or assembly due to deterioration from exposures in any test, or formation of any undesirable condition which cannot be remedied by routing maintenance procedures shall be cause for rejection.

4.4.5.1 LOW TEMPERATURE.- The stand shall be subjected to low temperature in accordance with procedure II, except the storage temperature shall be -65°F and the operating temperature -20°F.

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4.2.2.2.2 INDIVIDUAL TESTS MAY CONTINUE.- For production reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of the entire lot shall not be made until it is determined that the lot meets all the requirements of the specification.

4.2.3 DEFECTS IN ITEMS ALREADY ACCEPTED.- The investigation of a test failure could indicate that defects may exist in stands already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.3 TEST CONDITIONS

4.3.1 STANDARD ATMOSPHERIC CONDITIONS.- Unless otherwise specified, the test shall be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 73°F).

4.3.2 TEST PERIODS.- Unless otherwise specified, any test shorter than 1 hour shall not be credited as test time.

4.3.3 APPARATUS AND INSTRUMENTS.- Apparatus and instruments used in conjunction with testing specified herein shall be of laboratory precision type and shall be carefully checked to ascertain that they have been installed in a satisfactory manner for durability and accurate operation and shall be calibrated at intervals properly spaced to assure laboratory efficiency.

4.3.4 FLUID FLOW.- The calibration and performance checks of the displaced fluid shall be determined by means of a calibrated flowmeter with certified accuracy of 1 percent of reading or better.

4.4 TEST METHODS

4.4.1 EXAMINATION OF PRODUCT.- The stand shall be inspected to determine compliance with the requirements specified herein with respect to parts, materials, workmanship, and marking.

4.4.2 OPERATIONAL CHECK.- The constant flow operation of the stand shall be checked at 500, 1500, 2500, and 3000 psi at each rate of flow of 2, 4, 6, and 10 gpm and 5,000 psi at 5 gpm with compensation turned off. This test shall be conducted with the reservoir in the system, with the reservoir bypassed, and with the suction restricted to create a suction of 16 inches Hg absolute pressure at the inlet to the high-pressure pump (or 18 inches Hg vacuum to the booster pump). The pressure gage shall be accurate to within 1 percent of full scale at all settings. The calibrated

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The test circuit shall include one AN6292-8-120 pressure hose, a restrictor valve to build up 3,000 psi pressure, one reservoir filled with not less than 40 gallons of hydraulic fluid conforming to MIL-O-5606, and one MS28741-16-120 return line. Prior to starting the test, the entire test circuit shall be contained within the cold chamber and stabilized to -20°F. The fluid in the reservoir shall not be heated prior to starting or during the test from any external source. After starting, the flow and pressure shall be slowly increased to the rated output of 10 gpm and 3,000 psi. The stand shall attain rated output within 15 minutes after starting and shall be operated at rated output for not less than 30 minutes. All controls, gages, meters, valves, systems, and accessories shall be checked for satisfactory operation while operating in the -20°F ambient temperature.

4.4.5.2 HIGH TEMPERATURE.- The stand shall be subjected to high temperature in accordance with procedure I. Upon completion of the exposure to high temperature the ambient temperature shall be lowered to 125°F. The constant flow operation of the stand shall be checked at 800, 1,500, and 3,000 psi at each rate of 3, 6, and 10 gpm. The pressure gage shall be accurate to within ±1 percent of full scale at all settings. The calibrated flow control handwheel settings shall be accurate to within ±1/2 gpm of actual flow. The temperature shall then be reduced to 100°F. After this test, the stand shall be inspected for leakage of fluid.

4.4.5.3 SALT SPRAY.- The stand shall be subjected to salt spray as specified in MIL-E-4970.

4.4.5.4 HUMIDITY.- The stand shall be subjected to humidity as specified in MIL-E-4970.

4.4.5.5 RAIN.- The stand shall be subjected to rain as specified in MIL-E-4970. The stand shall be positioned on the test chamber floor in its normal operating position. After exposure to rain for 1 hour and while still in the rain test, the stand shall be started and operated at 10 gpm and 3,000 psi for a period of 30 minutes. Failure of the stand to start or operate satisfactorily during the rain test shall be cause for rejection. Upon completion of the rain test, all surfaces which might collect moisture shall be inspected to determine adequacy of drainage facilities and any collection of water in excess of 1/8-inch deep or which could be detrimental to the performance of the test stand shall be cause for rejection.

4.4.6 ENDURANCE.- The stand shall be connected through the hose specified in 3.9.16 to a test rig which shall include a reservoir, restrictors to create test pressures in each high pressure line, a restrictor to create a vacuum in the suction line, and a cooler to maintain fluid temperature at approximately 130°F. It shall then be subjected to a 300-hour endurance test as specified herein. The

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ambient temperature shall be maintained at 75°F, or higher, during the endurance tests. Reading shall be recorded every hour of all gages, meters, and instruments of the stand and output fluid flow, temperature, pressure and inlet fluid temperature, pressure, motor volts and amperes shall be measured as specified in 4.3 and recorded. Test procedures shall be as follows:

a. Conduct the tests specified in 4.4.2.

b. Fifty hours operation at 10 gpm and 3,000 psi. Pressure out through the 1/2-inch outlet and return through the 1-inch inlet (reservoir bypassed, 3/4-inch outlet idle). Continuous return pressure shall be 100 psig for this test.

c. Fifty hours operation at 5 gpm and 5,000 psi. Pressure out through the 1/2-inch outlet and return to reservoir through the 3/4-inch outlet for one-half of test; reverse for remainder of test. (Reservoir valve open, 1-inch inlet idle.)

d. Fifty hours operation at 10 gpm and 3,000 psi with suction pressure to the high-pressure pump inlet (after the suction filter) reduced to 16 inches Hg absolute pressure. Pressure out through the 1/2-inch outlet and return through the 1-inch inlet (reservoir bypassed, 3/4-inch outlet idle). If the boost pump is used to charge the high-pressure pump, the suction pressure before the boost pump shall be held at 18 inches Hg vacuum instead of the setting to the main pump specified above.

e. Fifty hours operation at 10 gpm and 1,500 psi. Alternate pressure out every hour between the 1/2-inch outlet and 3/4-inch outlet and return through the 1-inch suction inlet (reservoir bypassed).

f. Fifty hours operation at 2 gpm and 3,000 psi. Pressure out the 1/2-inch outlet, return through the 3/4-inch outlet to the reservoir (1-inch inlet idle, reservoir valve open).

g. Fifty hours operation at 5 gpm and 800 psi. Pressure out through the 3/4-inch outlet, return through the 1/2-inch outlet to the reservoir (1-inch inlet idle, reservoir valve open).

h. Conduct the tests specified in 4.4.2. Comparison of this data with data recorded under 4.4.6a shall indicate that no failure or undue wear has resulted from the endurance tests.

i. Operate the stand at 10 gpm at 3,000 psi with flow out the 1/2-inch outlet and returning into the 1-inch suction inlet. Vary the suction pressure from 16 inches Hg absolute to 110 psig.

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Failure of the stand to deliver rated flow or loss of hydraulic fluid from the system shall be cause for rejection.

4.4.7 ELECTRICAL EQUIPMENT TESTS.- The electrical system shall be inspected to determine compliance with 3.8.2. During the last hour of the full capacity run specified in 4.4.6, the temperature rise in the motor windings shall be determined by the resistance method and if the temperature rise exceeds the safe temperature limits for the class insulation used as specified in Standard C-50, it shall be cause for rejection. The motors shall be guaranteed by the motor manufacturer for safe temperature rise at 6,000 feet pressure altitude and full load and 104°F ambient.

4.4.7.1 Operation of the thermoswitch shall be checked at least twice and shall shut off the pump motor when the fluid to the high-pressure pump reaches not less than 158°F, or not more than 162°F.

4.4.7.2 The pump motor shall be overloaded to 125 percent of full load current and sustained until overload relays trip and shut the motor off. The overload relays shall trip before any damage is caused to the motor and in no case shall take longer than 15 minutes to open the main contactor. Ambient temperature for motor and control shall be 104° ± 9°F.

4.4.7.3 The stand shall be operated for 2 hours at 10 gpm and 3,000 psi with phase-to-phase voltage measured at the plug end of the electric cable reduced to 198 to 200V. The temperatures rise in the motor windings then shall be determined by the resistance method and shall not exceed the safe temperature limits for the class of insulation used. Ambient air temperature shall be held at 104°F, or over, during this test.

4.4.7.4 The stand shall be reconnected for 440V operation as specified in 3.8.2 and operated for a period of 2 hours in 104°F, or over, ambient temperature. The temperature rise in the motor windings shall then be determined by the resistance method and shall not exceed the safe temperature limits for the class of insulation used.

4.4.8 RADIO INTERFERENCE.- The stand shall be tested as a complete unit for radiated radio interference suppression. The test shall be conducted in accordance with MIL-I-6181 for radiated interference, except that whenever a shielded room is not readily available, a test location shall be selected in lieu of a shielded room where the inherent background level and the area background level are within the limits specified in MIL-I-6181. The meter

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and test specimens shall have a common ground potential in substitution for the ground plane and the antenna shall be limited to 6 feet from the perimeter of the test specimen.

4.4.9 NOISE LEVEL.- The noise level of the stand shall be measured 10 feet from the perimeter of the test specimen and shall not exceed the following noise level. Measurements shall be taken out-of-doors on a paved surface from at least eight different directions, 45° apart, in a horizontal plane parallel to the trailer bed and passing through the centerline of the high-pressure-pump drive shaft. During this test, the stand shall be operating at full rated load.

Cycles Per Second	Not Exceeding
20 to 75	106 db
75 to 150	98 db
150 to 300	90 db
300 to 600	83 db
600 to 1,200	78 db
1,200 to 2,400	75 db
2,400 to 4,800	72 db
4,800 to 10,000	71 db

4.4.10 OPERATION IN THE TILTED POSITION.- The stand shall rest on a platform and shall be kept from rolling by the parking brakes. The platform supporting the stand shall be tilted in four different directions, 90° apart, on an 8-1/2° incline with the true horizontal plane. The stand shall be started and operated at 10 gpm and 3,000 psi for 1 hour in each position. The stand shall perform satisfactorily in every respect. The parking brakes shall prevent the stand from moving on the platform. While tilted in each of the eight positions, 20 gallons of fluid shall be pumped from the reservoir to check the capacity specified in 3.9.5.

4.4.11 A 200-pound weight shall be applied slowly and uniformly to an area 4.5 by 5 inches in at least six random positions on top of the stand housing, including a reading in the center of each panel. The stand housing shall not be permanently deformed in any way by the application of the weight.

4.4.12 The stand shall be loaded with 800 pounds of weight distributed so that 400 pounds rests uniformly over the top of the housing, and 400 pounds are supported by the trailer frame. The stand shall then be towed by a standard 1/4-ton, 4 by 4 military truck, Type M38 at a speed of 40 mph for a distance of 50 miles over improved roads and 10 miles at 10 mph over unimproved roads.

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The 800 pounds of weights shall be removed and the stand shall execute 10 full turns to the right and 10 full similar turns to the left at an average speed of 10 mph. The turning radius for these turns shall be the minimum attainable with the 1/4-ton truck specified. Upon completion of the road tests, the stand shall be subjected to the tests specified in 4.4.2 and 4.4.2.1.

4.4.13 The flow control valves shall be tested by connecting the 1/2- and 3/4-inch pressure outlets to a double-acting hydraulic cylinder. With the selector valve directing pressure first to the 1/2-inch outlet (with its control valve full open) and then set to one side of the hydraulic cylinder's piston, the control valve in the 3/4-inch outlet shall allow the piston to move to any position and to stop the piston travel without creep. The above procedure shall be reversed to check the control valve in the 1/2-inch outlet. The working pressure for this test shall be 5,000 psi. Failure of the valves to satisfactorily manipulate the flow to the hydraulic cylinder or to stop the piston in any position without creep shall be cause for rejection.

4.4.14 SERVICEABILITY.- The stand shall be inspected and evaluated from the standpoint of ease of maintenance, servicing, and operation. Particular note shall be made with regard to provisions to prevent accumulation of dirt, snow, ice, et cetera, which may hinder servicing and operation.

4.4.15 FINAL DISASSEMBLY AND INSPECTION.- The stand shall be disassembled to the maximum practical extent for a thorough examination of all parts to disclose failure, distortion, excessive wear, or other unsatisfactory conditions. If measurements were made to determine wear they shall be compared with similar measurements made during initial disassembly inspection. Stands not conforming to the requirements specified herein shall be rejected.

4.4.16 TIE DOWN AND LIFTING.- The stand shall be subjected to the loadings specified in MIL-A-8421. The stand shall be secured during this test by means of the lifting and tie down fittings. Upon completion of this test, the stand shall be inspected to determine adequacy of bracing, adequacy of tie down or lifting fittings installation, and security of mounting of all components. The stand shall not be required to be operational after this acceleration test.

4.5 PREPRODUCTION TESTING

4.5.1 PREPRODUCTION TEST SAMPLES.- One stand, as specified in the contract, shall be tested for design approval by the procuring activity or, when so specified in the contract, at the contractor's plant under the supervision of the procuring activity.

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4.5.2 Samples submitted to the procuring activity for test shall be accompanied by the applicable bill of material and manufacturer's drawings, complete operating instructions, and adequate spare gaskets, locking devices, seals, et cetera, to provide replacements which may be necessary for the complete disassemblies and reassemblies required during the testing specified herein. The operating instructions shall contain specific information regarding the nature and extent of running-in, fuels, and lubricants to be used in the various test conditions specified herein, special precautions necessary for operation of the stand, and procedures to be followed in operating the stand under adverse climatic conditions.

4.5.3 PRELIMINARY RUN-IN AND CALIBRATION.- Any preliminary run-in of the stand, as may be recommended by the manufacturer, shall be performed by the manufacturer prior to submission of the stand for preproduction or acceptance tests. A certified log of all preliminary running time, calibration data, et cetera, shall be submitted to the procuring activity for entry in the log book.

4.5.4 PREPRODUCTION TEST REPORT.- When performance of the preproduction test by an activity other than the procuring activity is authorized, the manufacturer shall submit to the procuring activity a test report in accordance with MIL-T-9107.

4.5.5 PREPRODUCTION TESTS.- The preproduction tests shall consist of all the tests described under 4.4.

5. PREPARATION FOR DELIVERY

5.1 PRESERVATION AND PACKAGING

5.1.1 LEVEL A.- Stands shall be preserved in accordance with MIL-P-116, method IIA and as follows:

5.1.1.1 The stand shall be rigidly mounted to skids by means of a wood cradle, or equivalent, which shall support the weight of the stand so that the wheels are held at least 1 inch off the floor of the exterior container. The skids shall be firmly bolted to the skids of the container.

5.1.1.2 Hose and electrical cable shall be coiled to the minimum safe diameter and securely bound in three places with cotton tape. The bound coils shall be securely fastened to the stand within the moisture-impervious barrier.

5.1.1.3 The drawbar may be detached from and strapped to the stand.

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5.1.1.4 The hydraulic system shall be serviced to full operating capacity and all gages turned to the off position. All vent openings of the hydraulic system shall be plugged and taped with tape conforming to JAN-P-127 to prevent leakage of fluid.

5.1.1.5 The tires shall be inflated to 10 pounds above recommended pressure.

5.1.2 LEVEL C.- Stands shall be preserved and packaged in accordance with the manufacturer's commercial practice.

5.2 PACKING

5.2.1 LEVEL A.- Stands preserved and packaged as specified in 5.1.2 shall be packed in exterior shipping containers conforming to JAN-P-104. As far as practical, exterior containers shall be of uniform shape and size and of minimum cube and tare consistent with the protection required. Containers shall be closed and strapped in accordance with the container specification or appendix thereto. Containers shall be provided with a case liner conforming to MIL-L-10547, and shall be sealed in accordance with the appendix thereto.

5.2.2 LEVEL B.- Stands preserved and packaged as specified in 5.1.1 or 5.1.2, as applicable, shall be packed in domestic exterior containers conforming to MIL-C-3769. Exterior containers shall be of minimum cube and tare consistent with the protection required. Insofar as is practicable, exterior containers shall be of uniform shape and size. Containers shall be closed and strapped in accordance with the container specification or appendix thereto.

5.2.3 LEVEL C.- Packages which require overpacking for acceptance by the carrier shall be packed in exterior shipping containers in a manner that will insure safe transportation, at the lowest rate, to the point of delivery. Containers shall meet Consolidated Freight Classification rules or regulations of other common carriers as applicable to the mode of transportation.

5.3 PHYSICAL PROTECTION.- Cushioning, blocking, bracing, and bolting as required shall be in accordance with JAN-P-100, except for domestic shipments, waterproofing requirements for cushioning materials and containers shall be waived. The drop tests of JAN-P-100 shall be waived when preservation, packaging, and packing of the stand is for immediate use or when the drop tests of MIL-P-116 are applicable.

5.4 MARKING.- Interior and exterior containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be as follows:

Test Stand, Hydraulic System, Electric Motor Driven,
Type D-6A

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6. NOTES

6.1 USE.- The Type D-6A stand covered by this specification is intended for use in ground checking and maintenance of aircraft hydraulic systems.

6.2 ORDERING DATA.- Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Level of preservation and packaging, and packing required

6.2.1 PREPRODUCTION TESTS.- One stand will be required as a preproduction sample to be subjected to the preproduction tests to determine compliance with the requirements of the specification. The point of inspection for these tests will be specified.

6.3 DEFINITIONS.- For the purposes of this specification the following definitions shall apply:

6.3.1 IMPROVED ROADS.- An improved road is a road of level concrete, macadam, steel landing mats, or smooth, hard-packed dirt

6.3.2 UNIMPROVED ROADS.- An unimproved road is a road consisting of level-to-rolling gravel trails.

6.4 HAZARDS GUARANTEE.- The bidder will submit proof to the procuring activity that the electrical system specified in 3.10 which he proposes to supply under this specification conforms to standards of the Underwriters' Laboratories, Inc. as regards fir and casualty hazards. The label or listing of the Underwriters' Laboratories, Inc. will be accepted as evidence of conforming to the specification requirements.

6.4.1 In lieu of the label or listing, the bidder may submit independent proof satisfactory to the procuring activity that this electrical system conforms to the published standards, including methods of tests, of the Underwriters' Laboratories, Inc. Compliance with the fire and casualty hazards requirements does not absolve the bidder from complete compliance with the other requirements of this specification.

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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